Agile Board Reflection

HR Retention Analytics Consulting Proposal
Alpha Manufacturing Solutions

Rationale for Agile Decomposition

Our agile board was intentionally structured into six epics to mirror the progression of a real-world analytics consulting engagement while aligning directly to the original RFP requirements, including predictive capability, dashboard delivery, and structured governance. Although a complete workforce dataset was provided for the purposes of the course, the RFP emphasized SAP integration, governance standards, predictive modeling, and executive-ready dashboards. Our decomposition reflects those strategic priorities while adapting appropriately to the academic scope of working with a pre-assembled dataset.

Epic 1 – Governance & KPI Framework

We began with governance to ensure analytical alignment before conducting any modeling or dashboard work. Defining attrition logic, KPI measurement standards, and reporting cadence early to prevent inconsistencies later in the project. This reflects real consulting practices that analytics must be built on clearly defined business logic.

Even though we did not perform actual SAP integration, we conceptually incorporated governance structures that would be required in a real SAP-integrated environment, such as standardized retention definitions and documentation of measurement logic.

This sequencing ensures that downstream modeling and dashboards reflect agreed-upon definitions rather than ad hoc calculations.

Epic 2 – Data Foundation & Quality

Although the dataset was provided, we included a preparation epic to simulate real-world analytical rigor. This included:

- Standardizing turnover logic
- Defining validation rules
- Engineering modeling-ready features

Separating preparation from analysis reinforces reproducibility and structured workflow. It also aligns with consulting best practices, where analytical credibility depends on documented preprocessing steps.

Epic 3 – Descriptive & Statistical Analysis

This epic addresses the RFP requirement to identify where attrition occurs and why. By analyzing attrition across department, tenure, and role, and by quantifying statistically significant drivers, we ensure that predictive modeling was grounded in validated insight rather than being treated as a black box. This structure mirrors real consulting engagements, where stakeholders first require explanation before accepting predictive outputs.

Epic 4 – Predictive Modeling & Explainability

The RFP emphasized predictive capability and leadership confidence. Therefore, we created a dedicated modeling epic that includes:

- Baseline logistic regression
- Tree-based model comparison
- Hyperparameter tuning

- Model evaluation (AUC, precision, recall)
- Explainability framework
- Segment-level bias consistency checks

Separating model development from descriptive analytics allowed us to emphasize transparency and validation. This decomposition reinforces the principle that predictive modeling should be explainable and interpretable in HR contexts.

Epic 5 – Dashboard Development

We created a separate dashboard epic rather than embedding visualization into earlier stages. This reflects the RFP requirement for an executive-ready, drill-down capable retention dashboard.

We structured dashboard development into:

- Executive KPI summary views
- Attrition trend and cohort analysis pages
- Predictive risk visualizations
- Drill-down functionality
- Mobile optimization

This separation ensures that analytical outputs were translated into stakeholder-ready tools rather than remaining purely technical artifacts.

Epic 6 – Strategic Synthesis & Knowledge Transfer

The final epic focuses on converting analytics into actionable retention strategy and documentation. This directly addresses the RFP's emphasis on implementation roadmap and adoption.

Deliverables included:

- Retention strategy roadmap
- Executive presentation deck
- Technical documentation
- Dashboard training materials

Including this epic ensures that the engagement concluded with strategic value rather than stopping at model delivery.

<u>Alignment with RFP Requirements</u>

Our decomposition addresses the core RFP elements as follows:

Predictive Capability: Dedicated modeling epic with validation and explainability.

Executive Dashboard: Separate dashboard development epic with drill-down and mobile functionality.

Governance & Standardization: Early KPI framework and methodology planning epic.

Strategic Roadmap: Final synthesis and knowledge transfer epic.

Although we did not perform live SAP ERP integration due to dataset constraints, our governance structure reflects the documentation and standardization processes that would accompany enterprise integration in a real engagement.

<u>Challenges Encountered</u>

1. Translating a Strategic Proposal into Agile Structure

One challenge was decomposing a high-level consulting proposal into granular agile components. The RFP was written strategically, whereas agile planning required breaking deliverables into epics, features, and task-level items with ownership and sequencing.

This required careful thought to avoid overlap and ensure each epic had a clear purpose.

2. Sequencing Analytical Dependencies

Predictive modeling depends on validated descriptive insights and standardized logic. Ensuring that modeling tasks were not prematurely prioritized will require deliberate sprint sequencing. Agile planning helps address this by organizing epics week-by-week, reinforcing logical dependencies.

3. Simulating Governance Without Full Enterprise Context

Because the dataset was pre-provided, we did not perform actual data extraction or SAP integration. However, we still needed to demonstrate governance rigor. We addressed this by formalizing KPI definitions, measurement logic, and documentation tasks within Epic 1.

4. Ensuring Model Transparency

HR analytics requires heightened sensitivity around fairness and interpretation. Even without protected demographic variables, we included segment-level performance consistency checks to avoid disproportionate misclassification across departments or tenure bands.

Agile allowed us to isolate modeling validation as its own feature rather than treating it as an afterthought.

5. Coordinating Team Ownership

With four team members and six epics, ensuring equitable task distribution while maintaining logical workflow required coordination. Assigning two tasks per epic per member ensured balanced contribution and accountability.

6. Github Project Board Integration

No one on our team had used Github before, so learning how to correctly and properly add features and everything we need to the board has been challenging and a learning curve for the entire team as we all try to figure out how to best use the board.

<u>Role of Agile in HR Analytics Consulting</u>

Agile methodology enhanced this project by:

- Breaking complex analytical work into manageable weekly milestones
- Allowing logical sequencing of governance → analysis → modeling → dashboard → strategy
- Encouraging ownership through assigned tasks
- Supporting iterative refinement of dashboards and model validation

In HR analytics consulting, where stakeholder trust and interpretability are critical, agile enables structured transparency and controlled progression rather than a single “final delivery” event.

<u>Screenshots</u>

DATA 6520 Group 4 Project Board

Project Board | + New view

Filter by keyword or by field

Todo 5

This item hasn't been started

- Project_Board #2
  EPIC 2 - Data Foundation & Quality
- Project_Board #3
  EPIC 3 - Retention Analytics & Root Cause Analysis
- Project_Board #4
  EPIC 4 - Predictive Modeling
- Project_Board #5
  EPIC 5 - Dashboard Development
- Project_Board #6
  EPIC 6 - Strategy, Training & Closeout

+ Add item

In Progress 7

This is actively being worked on

- Project_Board #1
  EPIC 1 - Discovery & Planning
  Feb 19, 2026 Feb 26, 2026
  2 / 4 50%
- Project_Board #9
  Andrew - Tasks for Epic 1
  0 / 2 0%
- Project_Board #10
  Aidan - Tasks for Epic 1
  0 / 2 0%
- Project_Board #16
  Define attrition definitions (voluntary/involuntary/tenure logic)
- Project_Board #17
  Draft analytic methodology plan
- Project_Board #19
  Define stakeholder reporting cadence

+ Add item

Done 6

This has been completed

- Project_Board #7
  Jeff - Tasks for Epic 1
  2 / 2 100%
- Project_Board #8
  Connor - Tasks for Epic 1
  2 / 2 100%
- Project_Board #11
  Build Risk Register
- Project_Board #12
  Outline Data Governance
- Project_Board #14
  Draft project charter (scope, KPIs, success metrics)
- Project_Board #15
  Define retention KPIs and measurement logic

+ Add item

EPIC 1 - Discovery & Planning #1

Open · 2 / 4 · hernon33/Project_Board Public

hernon33 opened last week · edited by jeffreyvlass · Edits · Owner

Feature 1.1 – Project Governance & KPI Framework

User Story:
As a project team, we need clearly defined KPIs and governance standards so that all downstream analysis is aligned.

Feature Leads: Jeff, Connor

Feature 1.2 – Analytical & Dashboard Planning

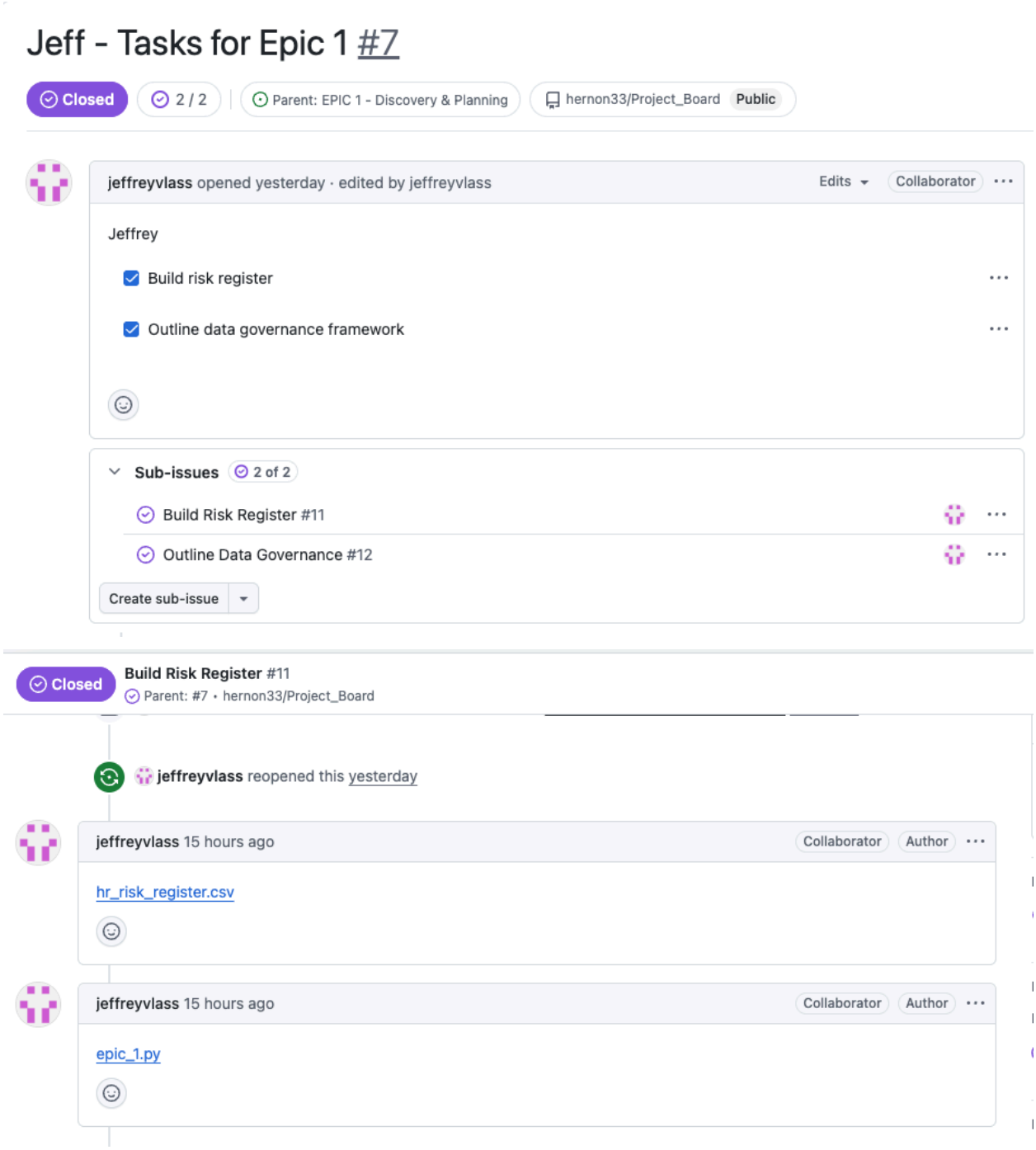
User Story:
As analysts, we need defined attrition logic and dashboard requirements so that analysis and visualization are structured correctly.

Feature Leads: Aidan, Andrew

Sub-issues 2 of 4

- Jeff - Tasks for Epic 1 #7 · 2 of 2
- Connor - Tasks for Epic 1 #8 · 2 of 2
- Andrew - Tasks for Epic 1 #9 · 0 of 2
- Aidan - Tasks for Epic 1 #10 · 0 of 2

Create sub-issue



Group Contribution Log

Connor – Governance framework tasks, descriptive attrition analysis, predictive explainability tasks, dashboard training materials

Jeffrey – KPI definition, statistical analysis tasks, baseline model development, executive dashboard build

Aidan – Analytical methodology planning, feature engineering, tree-based modeling, technical documentation

Andrew – Dashboard requirements document, visualization prototypes, mobile optimization, predictive dashboard integration

Each member will complete two tasks per epic and contribute across governance, analytics, modeling, and delivery components.